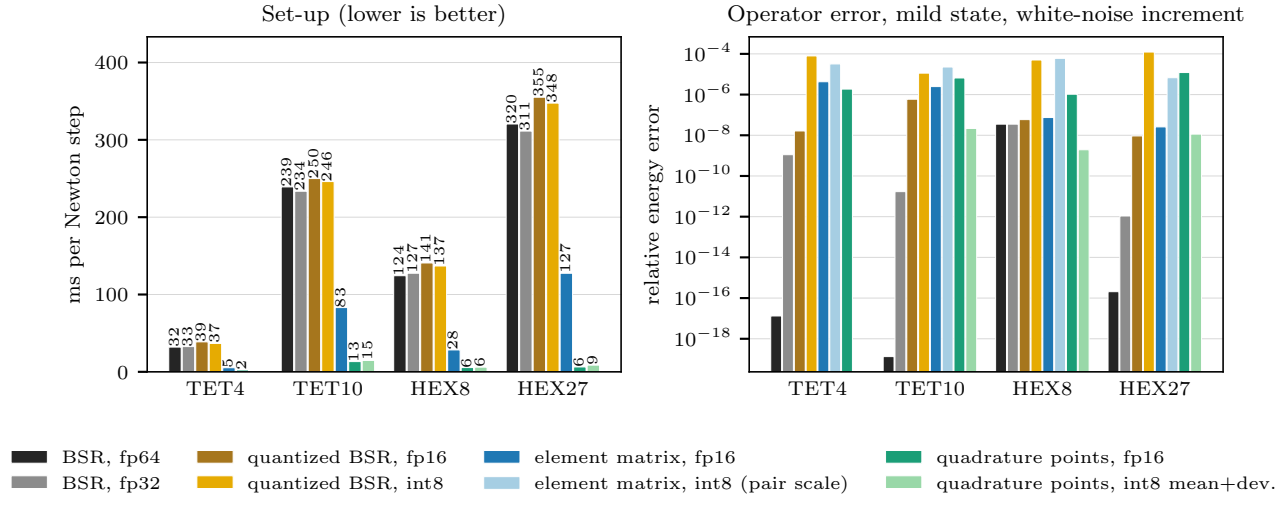

Set-up (lower is better)
ms per Newton step
400
300
200
100
0
32
33
39
37
5
2
239
234
250
246
83
13
15
124
127
141
137
28
6
6
320
311
355
348
127
6
9
TET4
TET10
HEX8
HEX27
Operator error, mild state, white-noise increment
relative energy error
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
10^{-18}
TET4
TET10
HEX8
HEX27
BSR, fp64
BSR, fp32
quantized BSR, fp16
quantized BSR, int8
element matrix, fp16
element matrix, int8 (pair scale)
quadrature points, fp16
quadrature points, int8 mean+dev.